

# Task-no.-5

Extract insights using visual and statistical exploration.

## Titanic Dataset: Exploratory Data Analysis Report

### Introduction

This report presents the findings from an exploratory data analysis (EDA) of the Titanic dataset, which contains information about the passengers aboard the RMS Titanic, including whether they survived the sinking of the ship on April 15, 1912. The analysis explores patterns and relationships in the data to better understand factors that influenced passenger survival.

### Dataset Overview

The Titanic dataset contains 891 records with 12 variables: - PassengerId: Unique identifier for each passenger - Survived: Survival status (0 = No, 1 = Yes) - Pclass: Passenger class (1 = 1st class, 2 = 2nd class, 3 = 3rd class) - Name: Passenger name - Sex: Gender of passenger - Age: Age of passenger (some values missing) - SibSp: Number of siblings/spouses aboard - Parch: Number of parents/children aboard - Ticket: Ticket number - Fare: Passenger fare - Cabin: Cabin number (many missing values) - Embarked: Port of embarkation (C = Cherbourg, Q = Queenstown, S = Southampton)

### Key Findings

**1. Missing Data - Age: 19.9% missing values - Cabin: 77.1% missing values - Embarked: 0.2% missing values (2 records)**

The high percentage of missing cabin data limits analysis of this feature, while the missing age data was handled appropriately in age-related analyses.

**2. Demographic Distribution - Gender: 64.8% male, 35.2% female - Class: 55.1% in 3rd class, 24.2% in 1st class, 20.7% in 2nd class - Age: Mean age of 29.7 years, with most passengers**

**between 20-40 years - Embarkation: 72.4% from Southampton, 18.9% from Cherbourg, 8.6% from Queenstown**

### **3. Survival Patterns**

**Overall Survival Rate - 38.4% (342 out of 891) passengers survived the disaster**

**Survival by Gender - Female: 74.2% survival rate - Male: 18.9% survival rate**

This dramatic difference confirms the "women first" evacuation policy was largely followed.

**Survival by Passenger Class - 1st Class: 62.9% survival rate - 2nd Class: 47.3% survival rate - 3rd Class: 24.2% survival rate**

The significant gradient in survival rates across classes suggests that socioeconomic status played a major role in determining who survived.

**Survival by Age - Children (0-12): Higher survival rates, especially for boys  
- Elderly (60+): Lower survival rates for men in this group - Age Groups by Gender: Females in all age groups had considerably higher survival rates than males**

**Survival by Family Size - Passengers in small family groups (2-4 members) had better survival rates (around 50%) - Passengers traveling alone had a lower survival rate (about 30%) - Passengers in large families (5+ members) had the lowest survival rates (below 20%)**

**Survival by Embarkation Port - Cherbourg: 55.4% survival rate - Queenstown: 38.9% survival rate - Southampton: 33.7% survival rate**

The higher survival rate for Cherbourg passengers is likely related to passenger class distribution, as more 1st class passengers embarked there.

### **4. Correlation Analysis - Strong negative correlation between passenger class and survival (-0.34) - Moderate positive correlation between fare and survival (0.26)**

- Strong negative correlation between being male and survival (-0.54) - Moderate correlation between family size and survival, with small families having positive correlation and large families having negative correlation

## Visualizations Insights

- **Gender and Class Interaction** The combination of gender and passenger class revealed particularly stark survival disparities: - 1st Class Women: 96.8% survival rate - 2nd Class Women: 92.1% survival rate - 3rd Class Women: 50.0% survival rate - 1st Class Men: 36.9% survival rate - 2nd Class Men: 15.7% survival rate - 3rd Class Men: 13.5% survival rate

**This pattern shows a clear hierarchy in evacuation priority, with women from higher classes having near-certain survival while men from lower classes had the lowest chances.**

- **Age, Gender, and Class Interaction** The three-way interaction between age, gender, and class revealed: - Female children in 1st and 2nd class had 100% survival rate - Male children in 1st and 2nd class had higher survival rates than adult males - Adult females in 3rd class had significantly lower survival rates than those in 1st and 2nd class - Male survival rates were consistently low across all age groups in 3rd class
- **Fare Distribution and Survival** - Passengers who paid higher fares had better survival rates - The fare distribution was highly skewed, with most passengers paying less than £50 - Within each class, survivors generally paid higher fares than non-survivors

## Conclusions

This exploratory data analysis of the Titanic dataset reveals several important insights about the factors that influenced passenger survival:

1. **Social Hierarchy:** The strong influence of passenger class on survival rates reflects the social stratification of the early 20th century. First-class passengers had access to better information, were likely closer to lifeboats, and may have received preferential treatment during evacuation.
2. **Gender Norms:** The "women and children first" maritime disaster protocol is clearly reflected in the data, with women having dramatically higher survival rates across all classes. This demonstrates how gender norms of the era played a critical role in determining survival outcomes.
3. **Family Dynamics:** The curvilinear relationship between family size and survival suggests that traveling with a small family was optimal for survival. Small families could stay together and help each other during evacuation, while very large families may have found it difficult to stay together in the chaos.

4. **Age Considerations:** Age influenced survival mainly in interaction with gender and class. Children generally had better chances, especially in higher classes, reflecting both the "women and children first" policy and the social hierarchy.
5. **Economic Factors:** The correlation between fare and survival rate, even within the same passenger class, suggests that economic resources played a role beyond the broad class categories, possibly influencing cabin location and access to information during the emergency. This analysis provides a data-driven glimpse into the human and social dynamics at play during one of history's most famous maritime disasters, showing how survival was influenced by a complex interplay of demographic, economic, and social factors.